جامعة الأمير سطام بن عبد العزيز

كلية هندسة و علوم الحاسب

Adding the configuration management module to the system

SE – 4111 Course Project

Submitted to: Dr. Mawal A. Mohammed

Prepared by:

S	ID	Name
1	441051251	Shahad Abdullah AlNoaisser
2	441051670	Rema Mohammed Aldawood
3	441051382	Ashwaq Rajeh Alotaibi

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Table of Contents

1. Requirements 3
2. Design 4
2.1 High-level design 4
2.2 Detailed design 4
2.3 User-interface design 5
3. Implementation 6
3.1 Development Environment 6
3.2 Code 16
3.3 User interface 22
4. Testing 23
5. Lessons learned 28

Important Notes:

1. **Font**: Times new romans
2. **Section titles** (size: 14; Bold)
3. **Section text** (size:12; space between lines 1.5; justification: fill justification)

Executive Summary

first, in the requirements we wrote the objectives and the features of our project ,then in the Design section, in the High-level design we explained the MVC architecture , in Detailed design we wrote the specification of set and get methods , also in User-interface design we added a hand draw of our implementation which is the output of our settings. in Implementation section , in Development Environment we described the steps that we did in phase 1, in the code we added the code of our singleton class and MVC , in User interface we added screenshot of our settings output. in Testing we added screenshots of the test along with its results. finally in Lessons learned we summarized what we learn from this project

1. Requirements

The goal of developing this system is to add the configuration management module to it and some of its features that it have only one instance (object) and accessor function which is global access to this instance and we can set certain values that are controlled from this instance and every component in the system can have access to it , also it have a lazy instantiation in the accessor function to improve the performance, also It's makes all constructors private so that they cannot be accessed from other classes .

Configuration management helps to build robust and stable systems through the use of tools that automatically manage and monitor updates to configuration data. and helps with project management, asset management and audit processes, as well as software development and debugging. It also provides a Quicker restoration of service and it Reduced risk of outages and security breaches through visibility and tracking of the changes to systems and it has Greater agility and faster problem resolution, enable to provide a higher quality of service and reduce software engineering costs by having detailed knowledge of all the elements of the configuration, avoiding wasteful duplication of the technology assets.

2. Design

2.1 High-level design

MVC Pattern stands for Model-View-Controller Pattern which is an architectural pattern that separates an application into three main logical components: the model, the view, and the controller. Each of these components are built to handle specific development aspects of an application and MVC mostly relates to the UI / interaction layer of an application.

- Model - The Model component corresponds to all the data-related logic that the user works with. This can represent either the data that is being transferred between the View and Controller components or any other business logic-related data. For example, Model represents an object or JAVA POJO carrying data. It can also have logic to update controller if its data changes
- View - View represents the visualization of the data that model contains.
- Controller - Controller acts on both model and view. It controls the data flow into model object and updates the view whenever data changes. It keeps view and model separate.

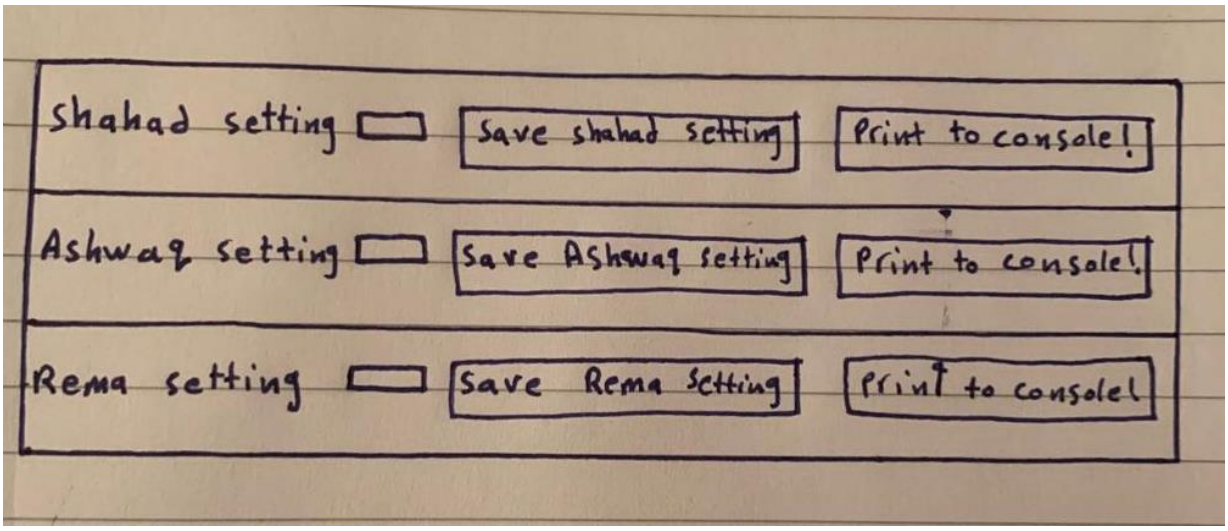
2.2 Detailed design

public void setShconfig(int con1) requires: con1 value to pass a value into the set function effects:to make sure the con1 is int
public int getShconfig() effects: return the shaconfig value

public void setAshconfig(int con2) requires: con2 value to pass a value into the set function effects:to make sure the con2 is int
public int getAshconfig() effects: return the Ashconfig value

public void setReconfig(int con3) requires: con3 value to pass a value into the set function effects:to make sure the con3 is int
public int getReconfig() effects: return the Reconfig value

2.3 User-interface design

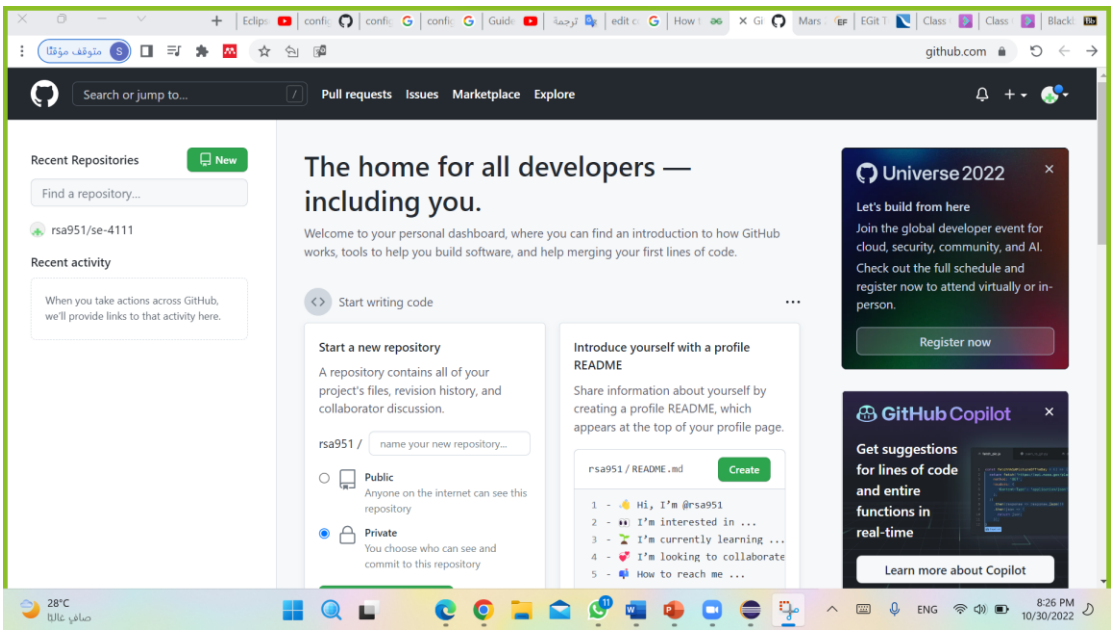


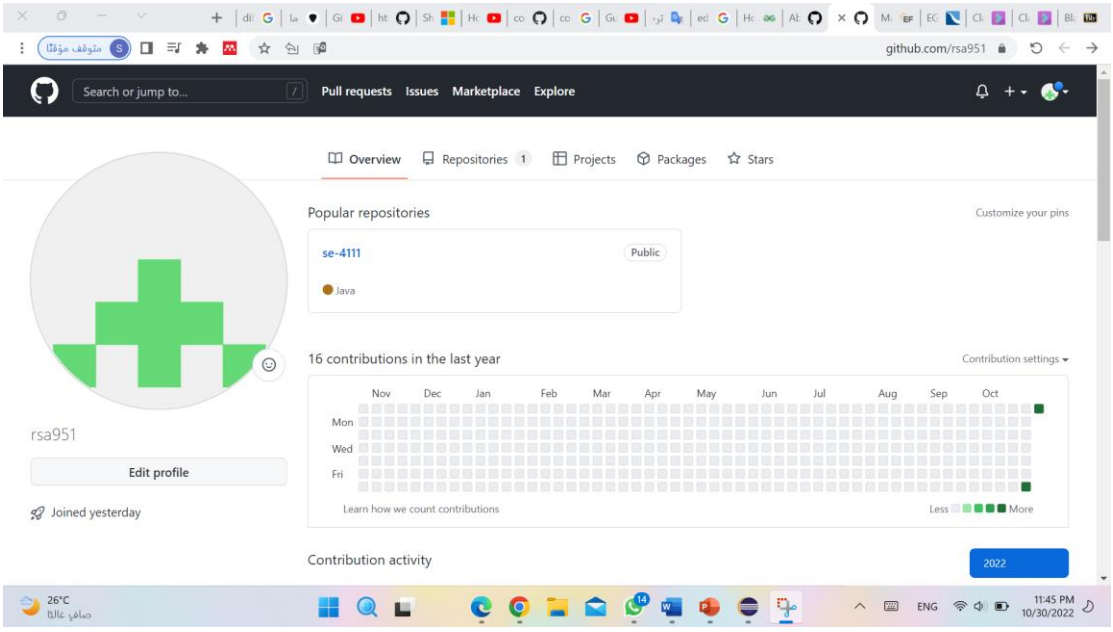
3. Implementation

3.1 Development Environment

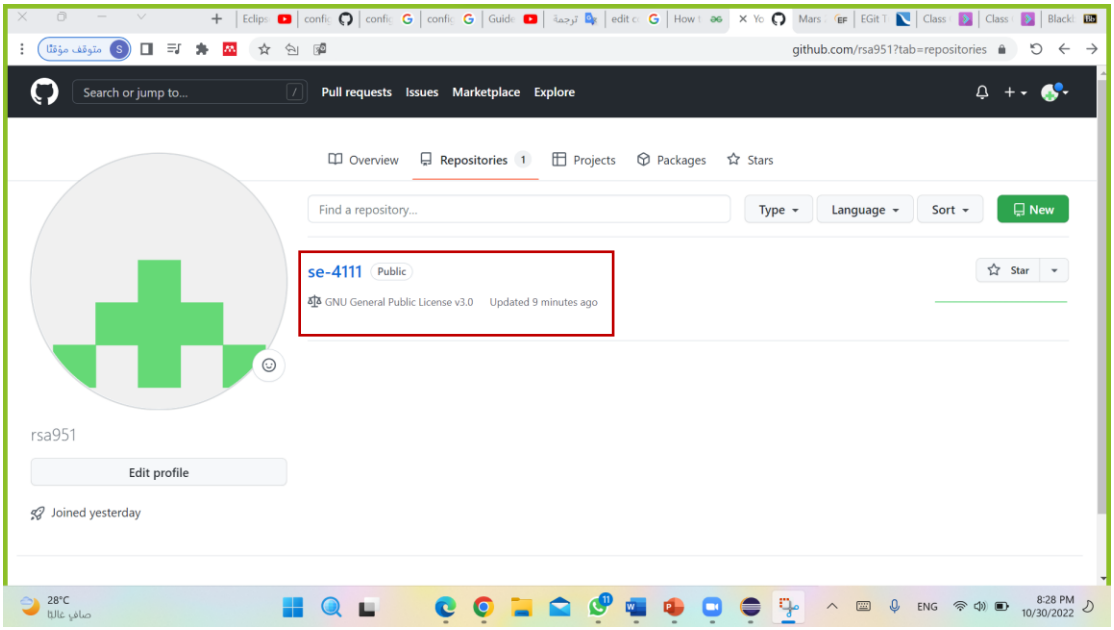
The Link to our repository: https://github.com/rsa951/se-4111

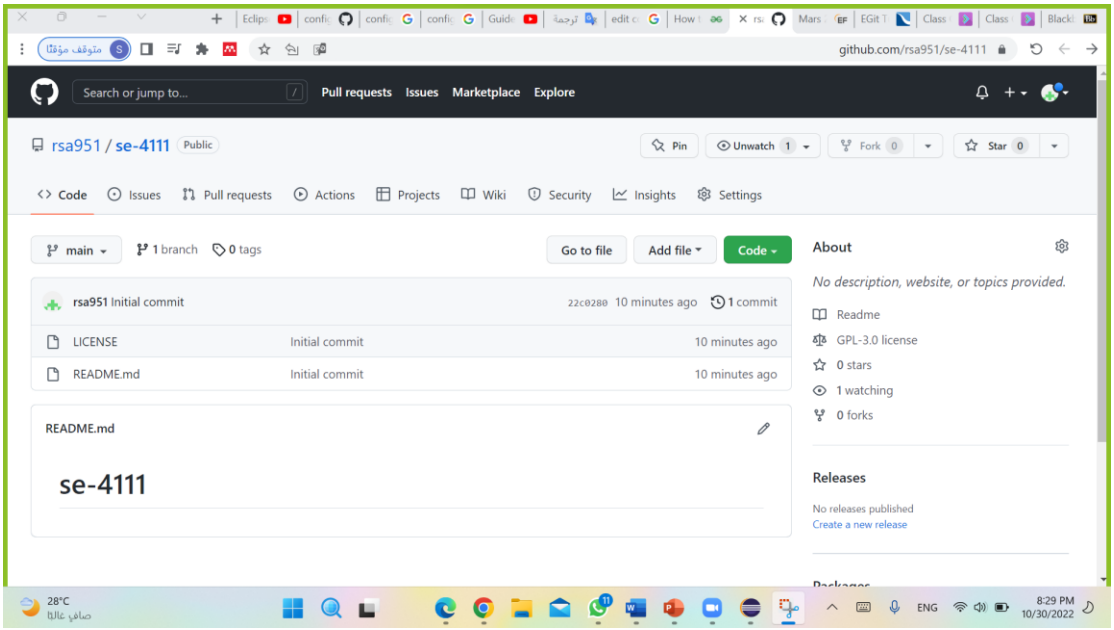
First, we created an account on GitHub so that all group member can participate in it





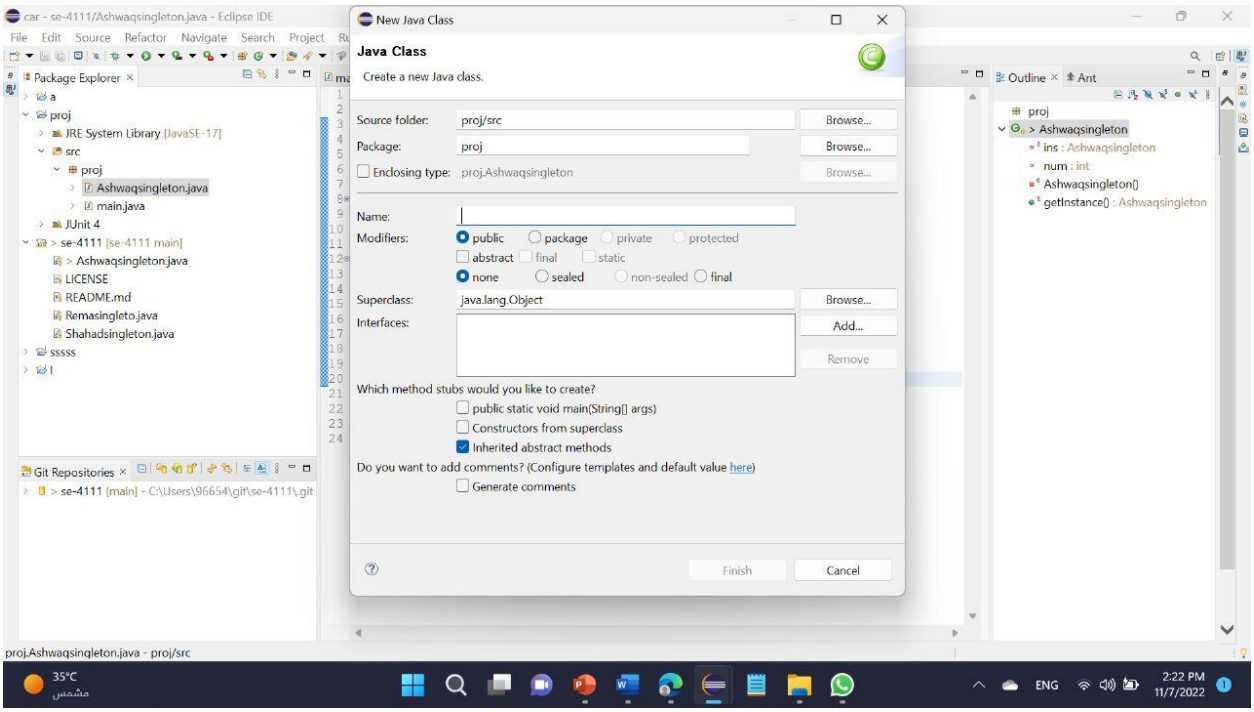
Second, we created a one repository in the created Github account and we named it “se-4111”

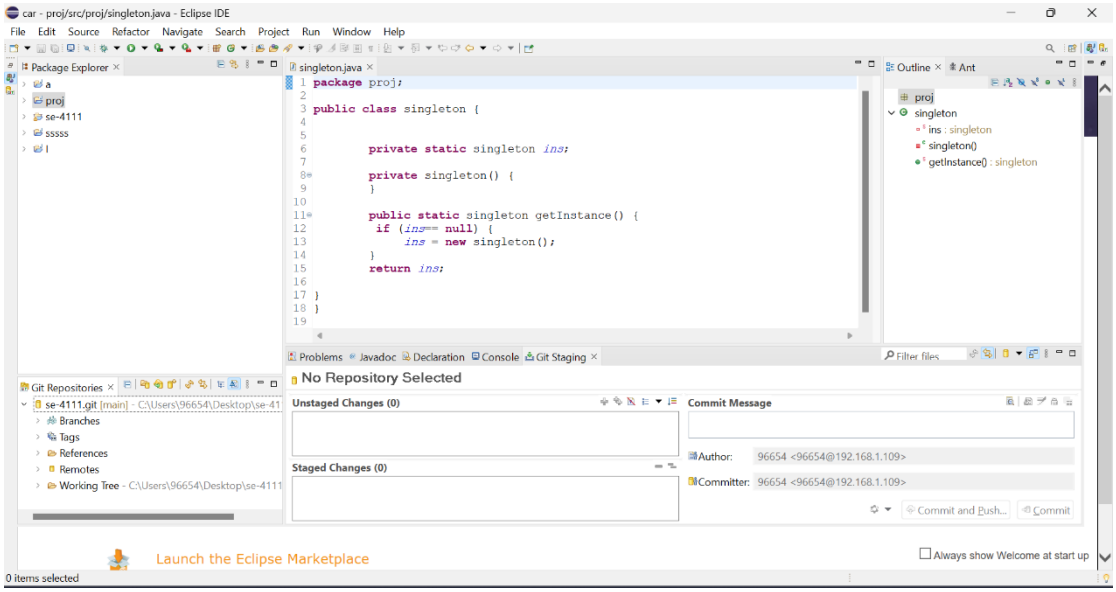




Third, we write a singleton class and then we put it in our repository, after that each member pulled the class and added her name in the singleton constructer.

Here are the steps for creating the singleton class

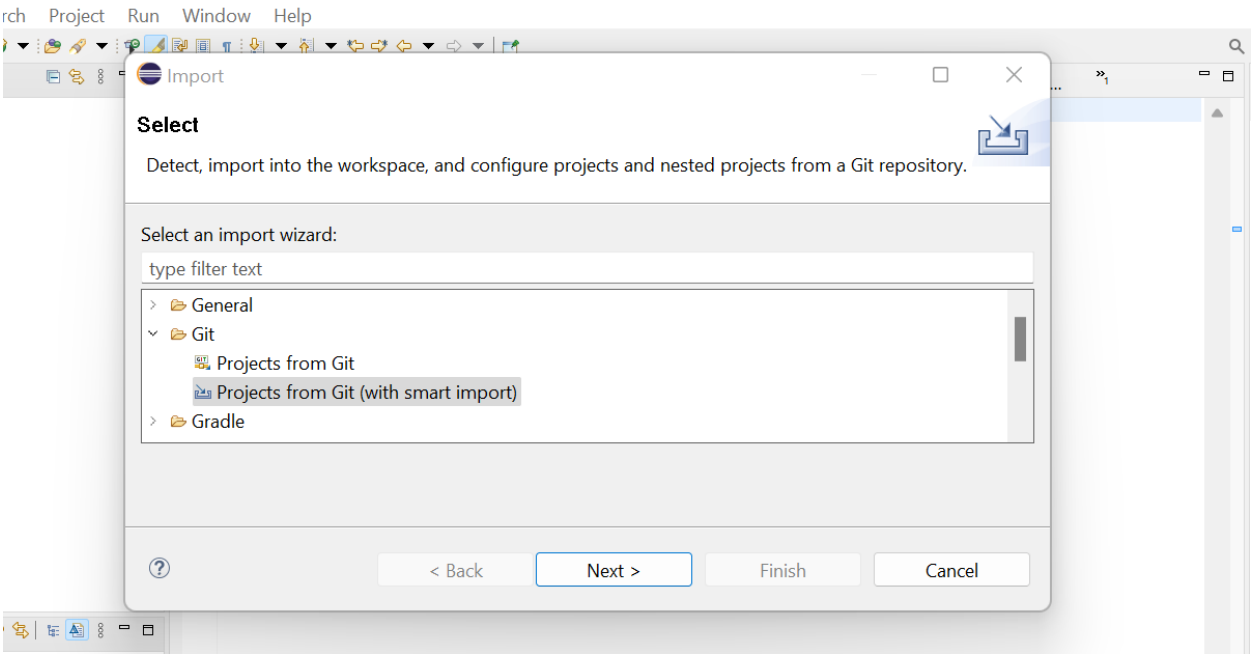


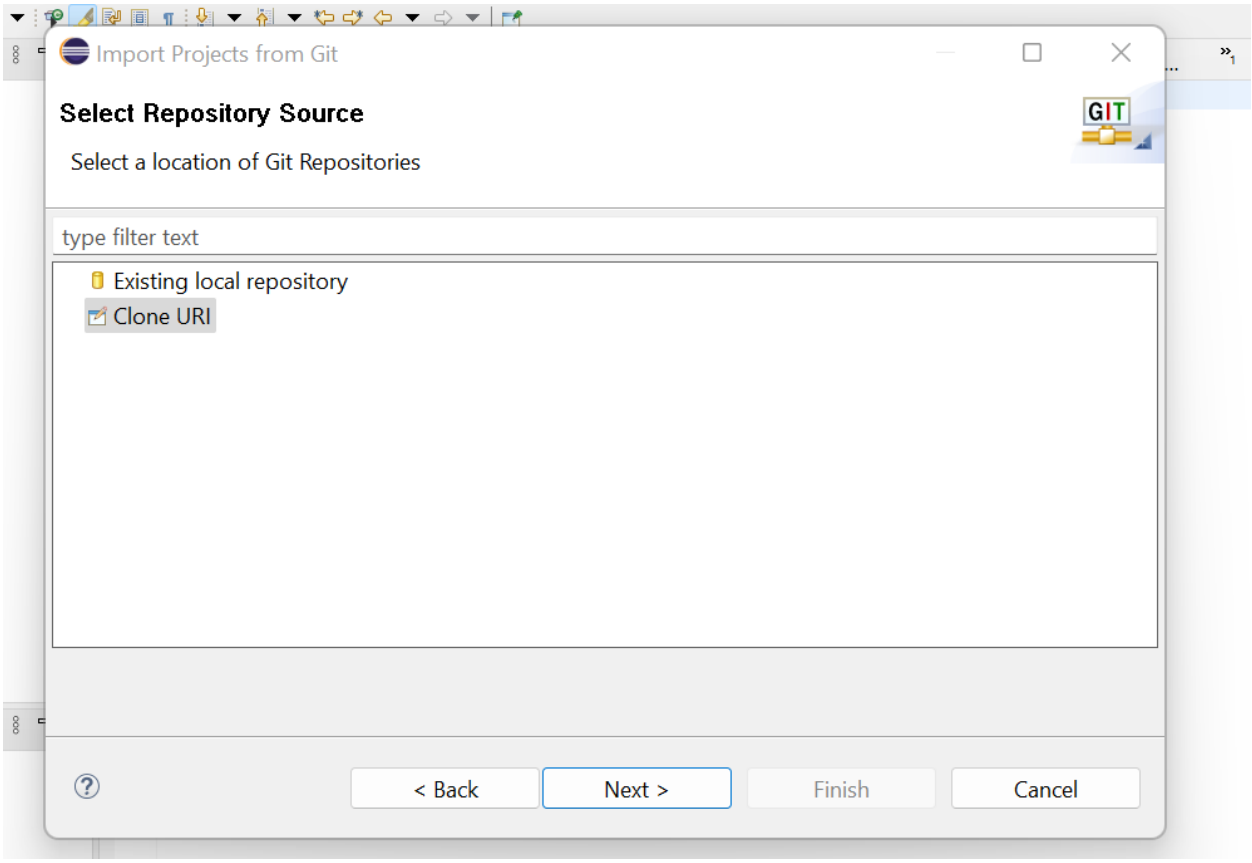


Then we put it in our repository, so that every member can pull the class and write her name.

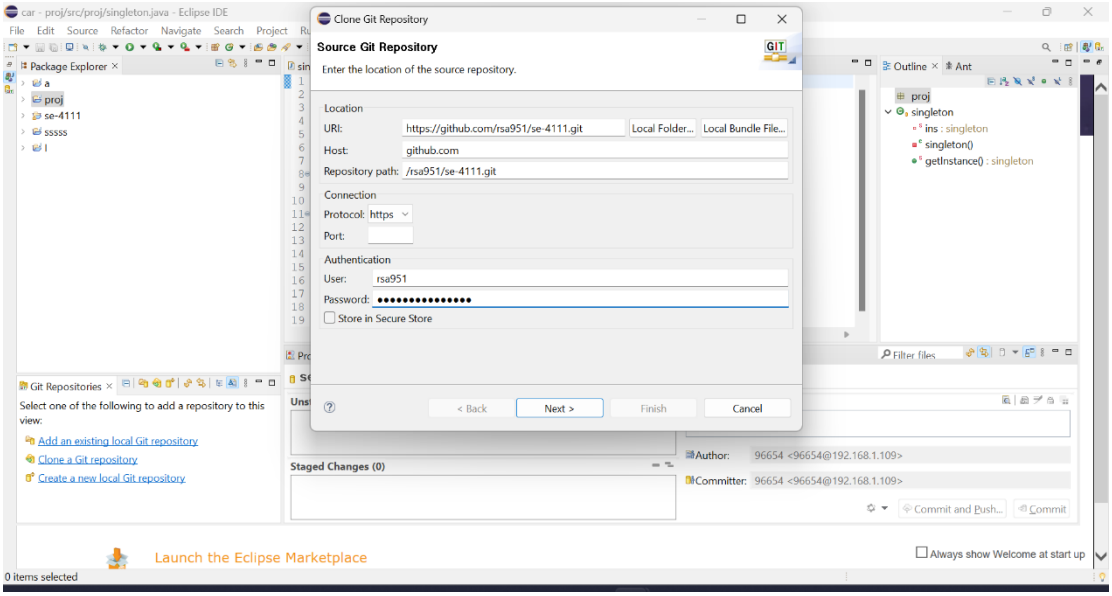
Here are the steps of importing the repository in our eclipse.

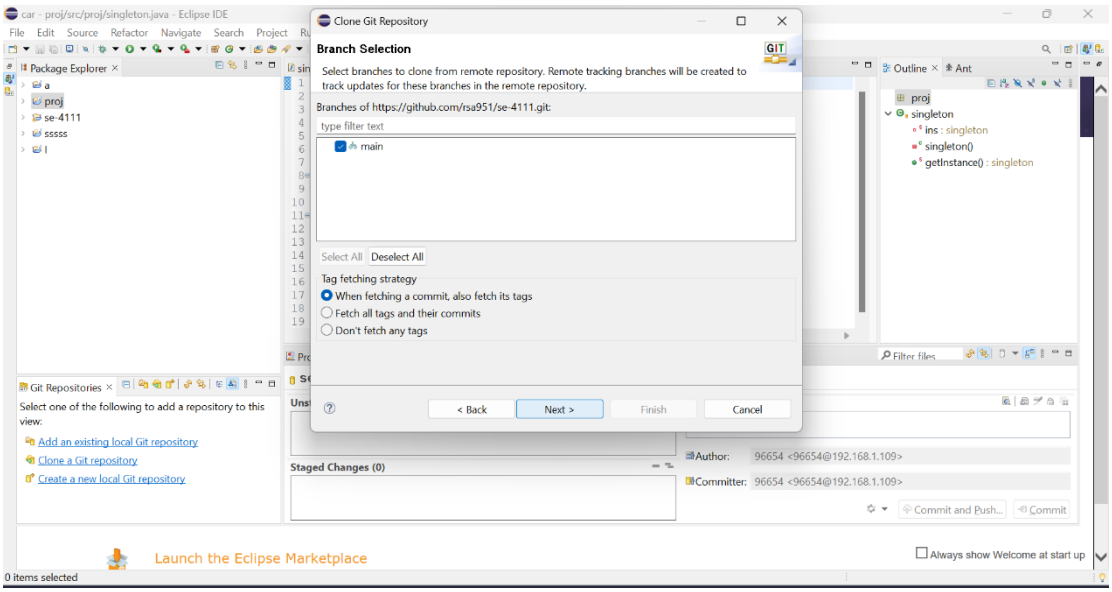
First, we went to file →import and then we chose the options as shown in these pictures

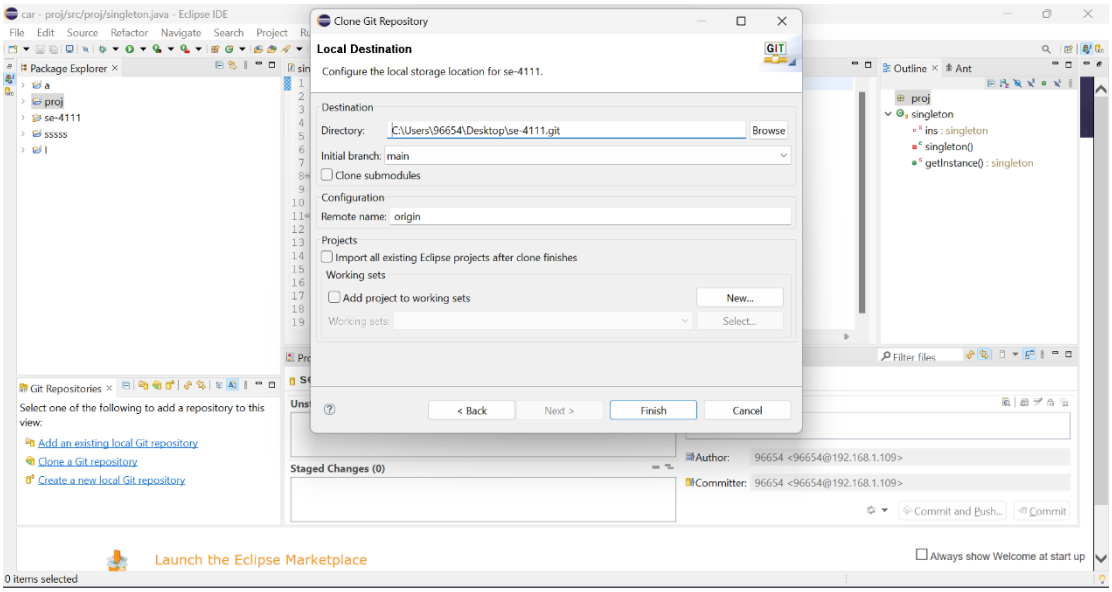




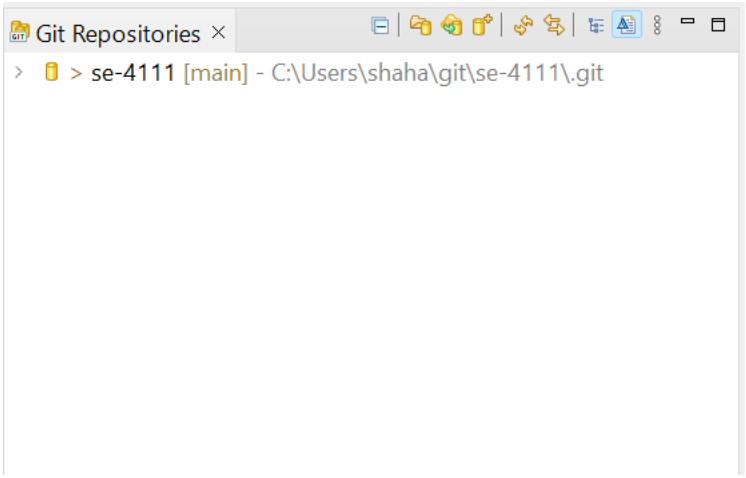
And here we put the link of our repository as well as user and password





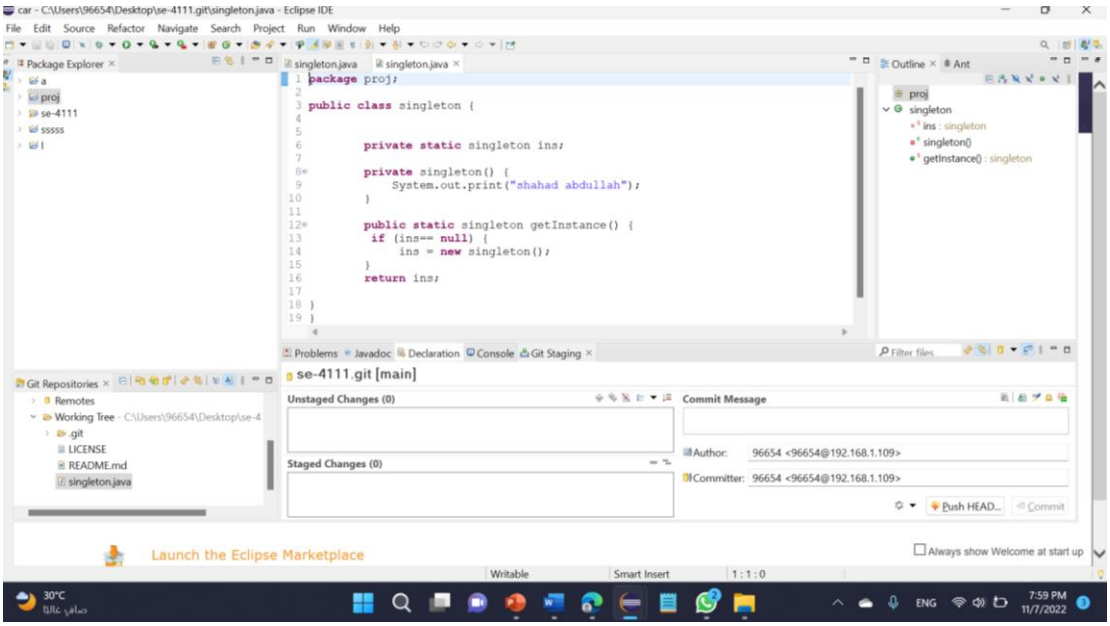


Finally, we get our repository

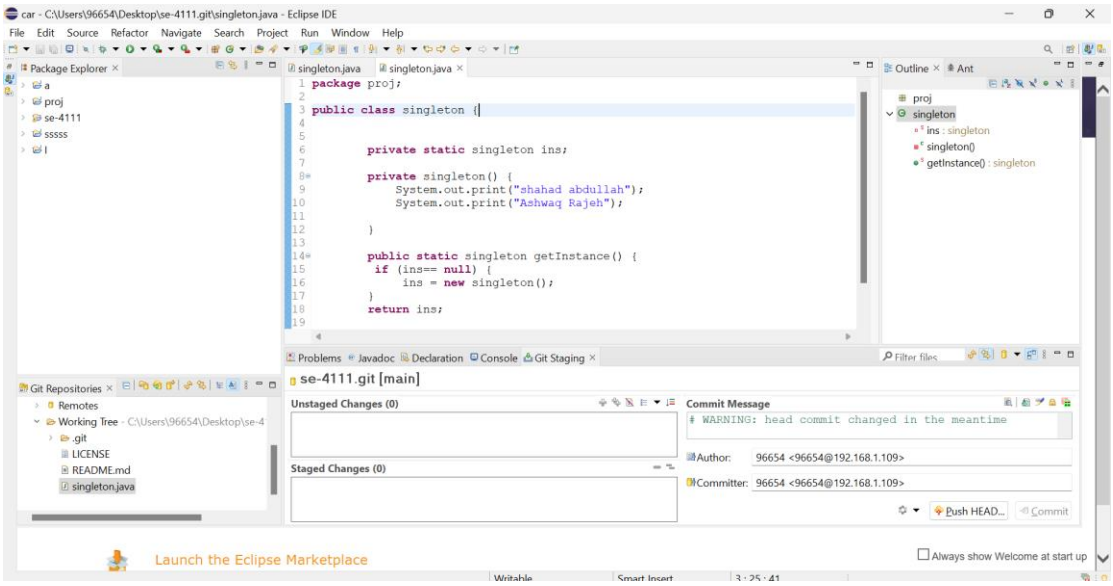


After that each member opened the singleton class file and wrote her name

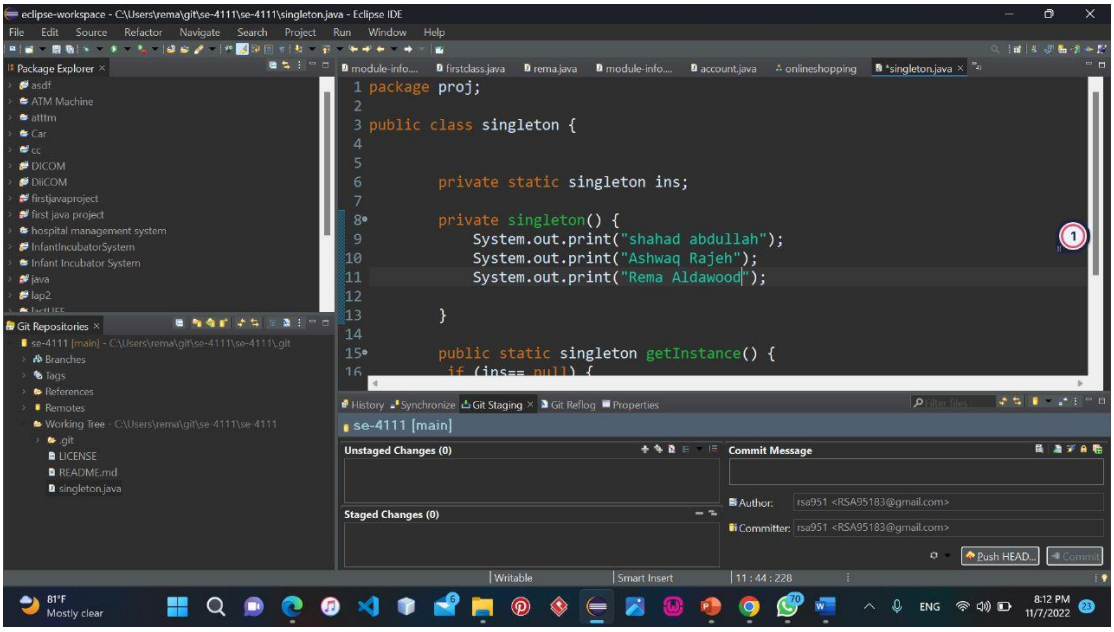
Shahad:



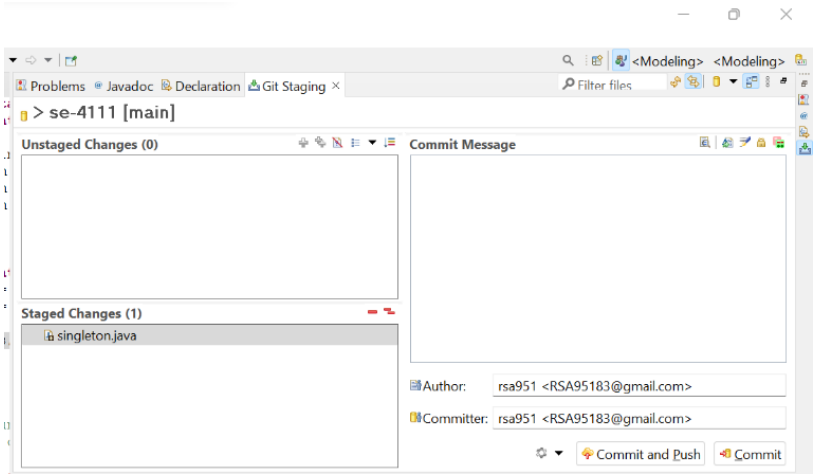
Ashawq

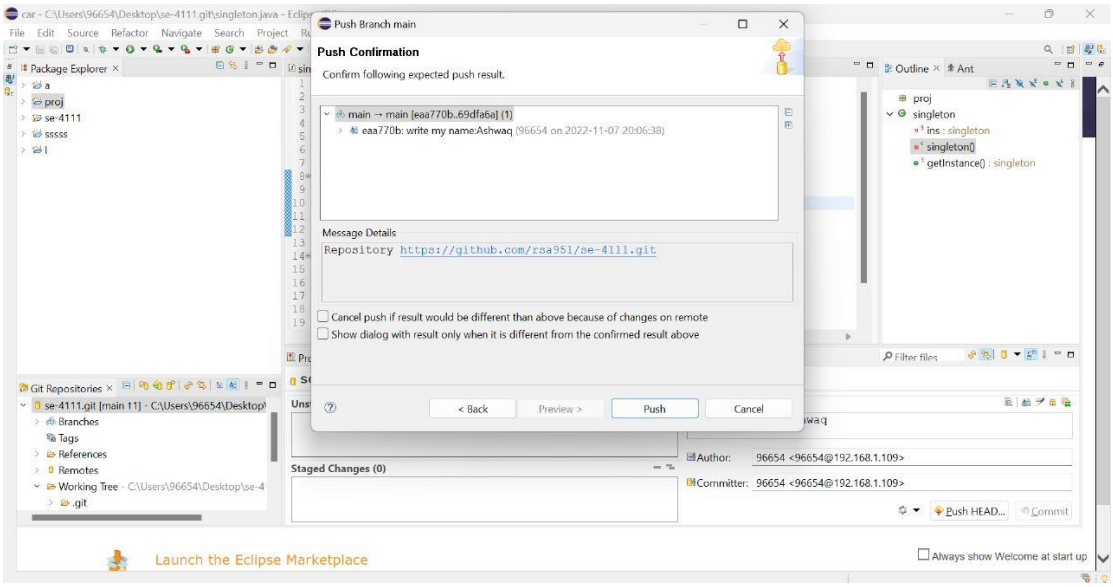


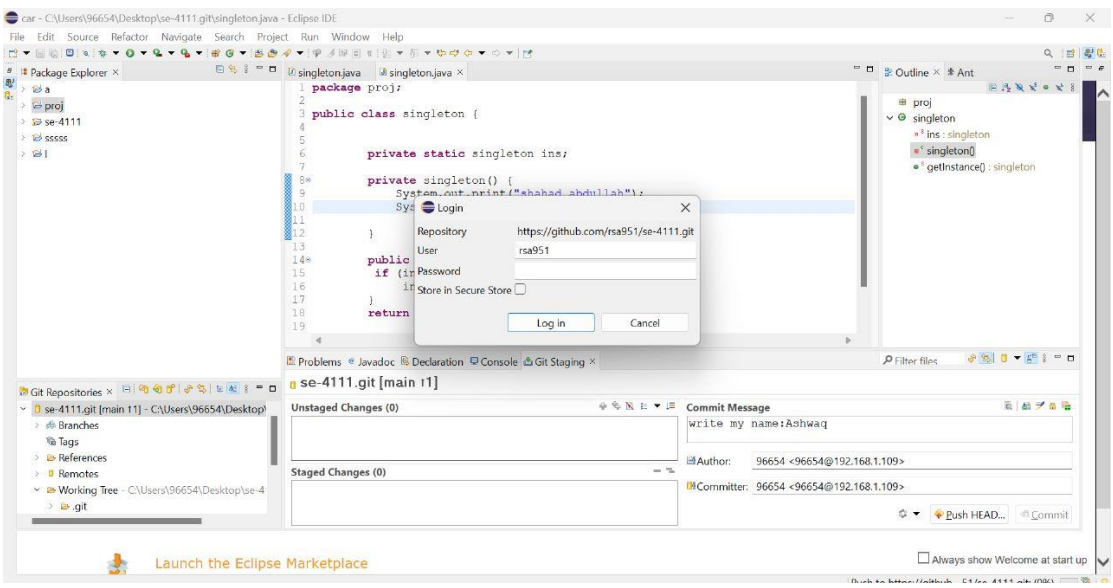
Rema:

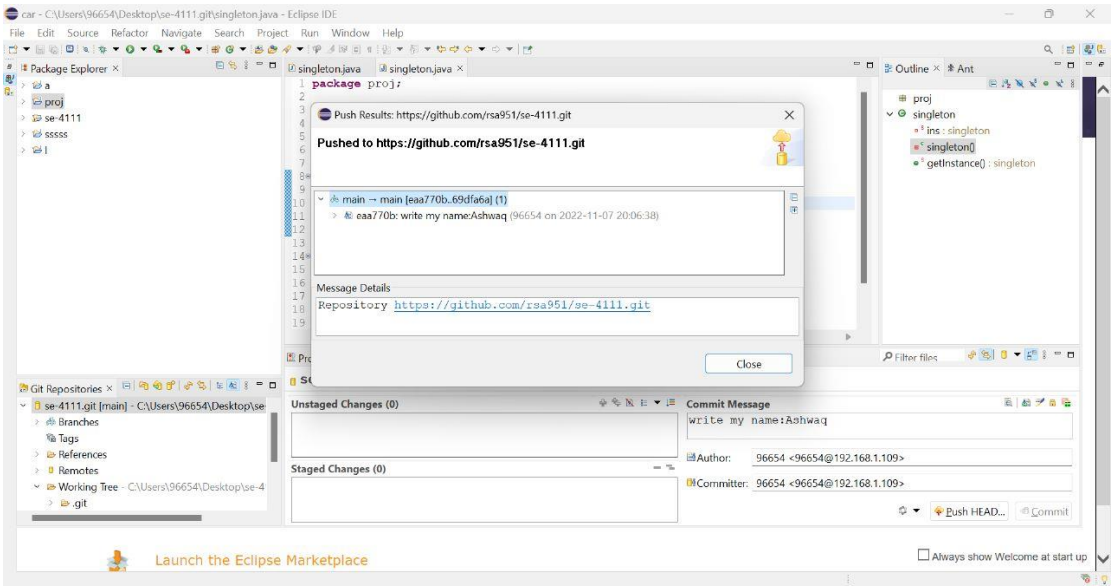


After that each member did the same steps as shown in the pictures to push their changes back into repository on GitHub

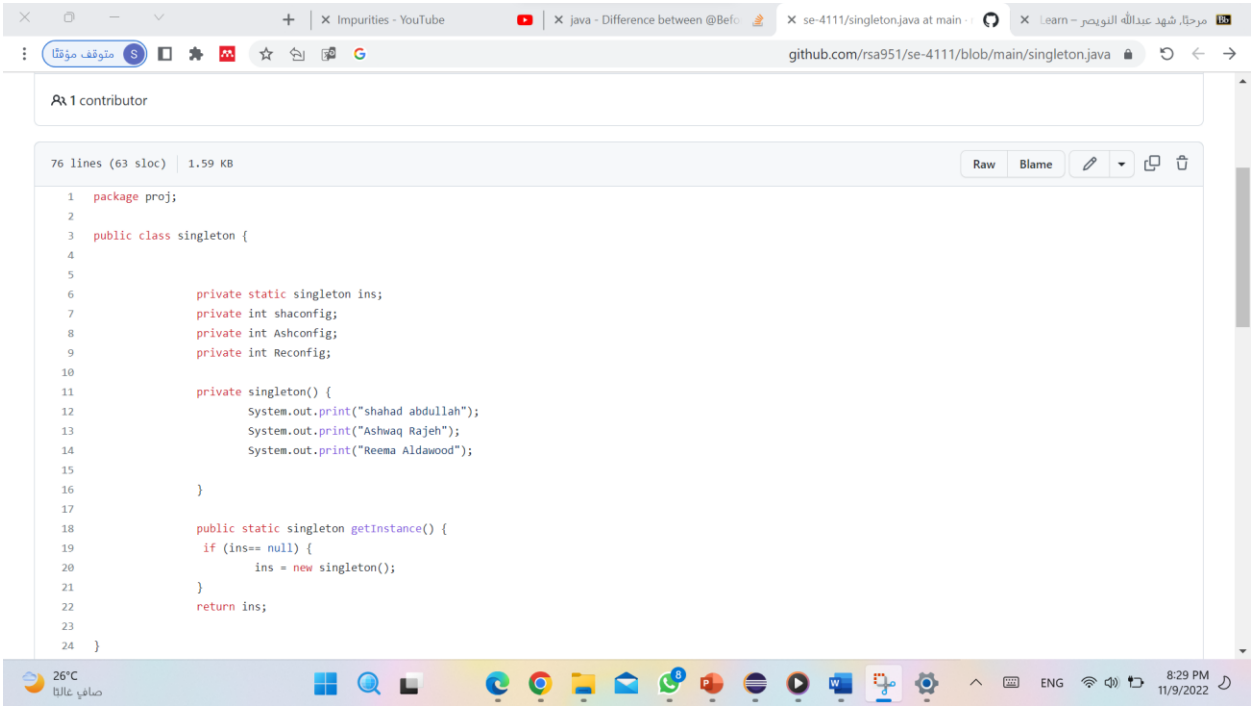



Push Branch main
Push Confirmation
Confirm following expected push result.
main → main [eaa770b..69dfa6a] (1)
eaa770b: write my name:Ashwaq (96654 on 2022-11-07 20:06:38)
Message Details
Repository https://github.com/rsa951/se-4111.git
Cancel push if result would be different than above because of changes on remote
Show dialog with result only when it is different from the confirmed result above
< Back
Preview >
Push
Cancel
Staged Changes (0)
Author: 96654 <96654@192.168.1.109>
Committer: 96654 <96654@192.168.1.109>
Push HEAD...
Commit
Launch the Eclipse Marketplace
Always show Welcome at start up


package proj;
public class singleton {
private static singleton ins;
private singleton() {
Login
Repository https://github.com/rsa951/se-4111.git
User rsa951
Password
Store in Secure Store
Log in
Cancel
se-4111.git [main ↑1]
Unstaged Changes (0)
Staged Changes (0)
Commit Message
write my name:Ashwaq
Author: 96654 <96654@192.168.1.109>
Committer: 96654 <96654@192.168.1.109>
Push HEAD...
Commit
Launch the Eclipse Marketplace
Always show Welcome at start up



And here is our singleton class including all the changes in repository on GitHub



3.2 Code

https://github.com/rsa951/se-4111/blob/main/singleton.java

this is the code of singleton class with its setters and getters and the specification of these two functions → phase 2

```
package proj;

public class singleton {
		private static singleton ins;
		private int shaconfig;
		private int Ashconfig;
		private int Reconfig;

		private singleton() {
			System.out.print("shahad abdullah");
			System.out.print("Ashwaq Rajeh");
			System.out.print("Reema Aldawood");   }

		public static singleton getInstance() {
		 if (ins== null) {
			 ins = new singleton(); }
		return ins;}

		/*
		 * this function is to set an int number to shaconfig
		 * @param con1 value to pass a value into the set function
		 */
		public void setShconfig(int con1) {
			assert(con1 % 1 == 0);
			shaconfig=con1;}
		/*
		 * this function is to get and return the shaconfig value
		 * @return shaconfig value
		 */
		public int getShconfig() {
		return shaconfig;}

		/*
		 * this function is to set an int number to Ashconfig
		 * @param con2 value to pass a value into the set function
		 */
		public void setAshconfig(int con2) {
			assert(con2 % 1 == 0);
			Ashconfig=con2;}
		/*
		 * this function is to get and return the Ashconfig value
		 * @return Ashconfig value
		 */
		public int getAshconfig() {
		return Ashconfig;}
		/*
		 * this function is to set an int number to Reconfig
		 * @param con3 value to pass a value into the set function
		 */
		public void setReconfig(int con3) {
			assert(con3 % 1 == 0);
```

```
                    Reconfig=con3;}

            /*
             * this function is to get and return the Reconfig value
             * @return Reconfig value
             */
            public int getReconfig() {
            return Reconfig;} }}
```

these are is the code of Settings Component ,Settings Controller , Settings Model and Settings View → phase 3

Settings Component:

```
public class SettingsComponent {
        public static void main(String[] args) {
         //  Assemble all the pieces of the MVC
 SettingsModel m = SettingsModel.getInstance();
         SettingsView v = new SettingsView();
         SettingsController c=new SettingsController(m, v);
         c.initControllerSh();
         c.initControllerASH();
         c.initControllerRE();  }  }
```

Settings Controller:

```
            public class SettingsController {
        private SettingsModel model;
        private SettingsView view;
   public SettingsController(SettingsModel m, SettingsView v) {
         model = m;
         view = v;
         initViewSh();
         initViewASH();
         initViewRE(); }
         //shahad
        public void initViewSh() {
                view.getSettingShTextfield().setText(Integer.toString(model.getShconfig()));}
         public void initControllerSh() {
                view.getSettingShSaveButton().addActionListener(e -> saveSettingOneSh());

                view.getSettingShPrintButton().addActionListener(e -> printToConsoleSh());}
        private void saveSettingOneSh() {
                model.setShconfig(Integer.parseInt(view.getSettingShTextfield().getText())); }
        private void printToConsoleSh() {
                System.out.println(model.getShconfig());}
         //Ashwaq
         public void initViewASH() {
                view.getSettingASHTextfield().setText(Integer.toString(model.getASHconfig()));}
         public void initControllerASH() {
                view.getSettingASHSaveButton().addActionListener(e -> saveSettingOneASH());
                view.getSettingASHPrintButton().addActionListener(e -> printToConsoleASH());}
        private void saveSettingOneASH() {
                model.setASHconfig(Integer.parseInt(view.getSettingASHTextfield().getText()));}
        private void printToConsoleASH() {
                System.out.println(model.getASHconfig());}
        //Rema
    public void initViewRE() {
```

```
                    view.getSettingRETextfield().setText(Integer.toString(model.getREconfig()));}
          public void initControllerRE() {
                    view.getSettingRESaveButton().addActionListener(e -> saveSettingOneRE());

                    view.getSettingREPrintButton().addActionListener(e -> printToConsoleRE());}
         private void saveSettingOneRE() {
                    model.setREconfig(Integer.parseInt(view.getSettingRETextfield().getText()));  }
         private void printToConsoleRE() {
                    System.out.println(model.getREconfig());}}
```

Settings Model:

```
public class SettingsModel {
          private static SettingsModel uniqueInstance;
//  other useful instance variables here
     private SettingsModel() {
                    }
                    private int shaconfig;
                    private int ASHconfig;
                    private int REconfig;
                    public static SettingsModel getInstance() {
                              if (uniqueInstance == null) {
                                        uniqueInstance = new SettingsModel();}
                              return uniqueInstance;}
          //shahad
                    public void setShconfig(int con) {
                              assert con>0;
                              shaconfig=con;}
                    public int getShconfig() {
                    return shaconfig;}
           //Ashwaq
      public void setASHconfig(int con) {
                              assert con>0;
                              ASHconfig=con;}
                    public int getASHconfig() {
                    return ASHconfig;}
      //Rema
      public void setREconfig(int con) {
                    assert con>0;
                    REconfig=con;}
            public int getREconfig() {
                                        return REconfig;}}
```

Settings View:

```
import java.awt.BorderLayout;import javax.swing.GroupLayout;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JTextField;
```

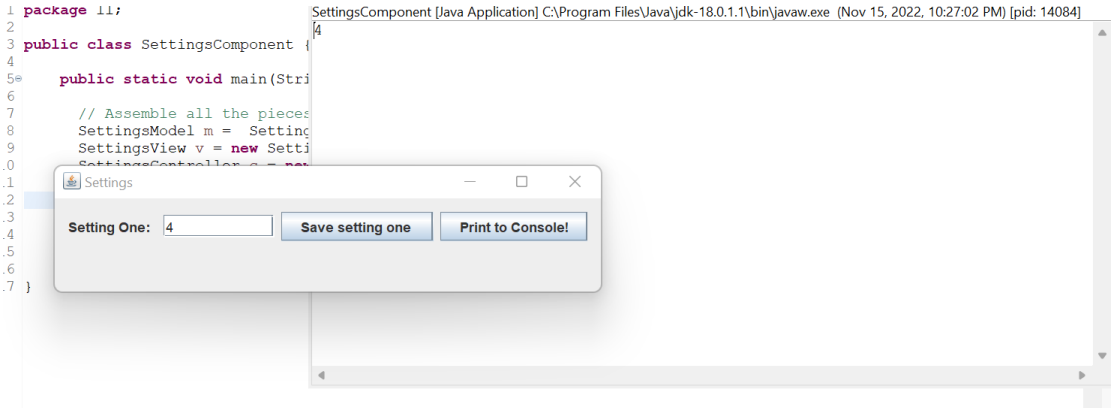
```
public class SettingsView {
        private JFrame frame;
         // shahad
   private JLabel settingShLabel;
        private JTextField settingShTextfield;
   private JButton settingShSaveButton;
        private JButton settingShPrintButton;
        //Ashwaq
          private JLabel settingASHLabel;
          private JTextField settingASHTextfield;
                private JButton settingASHSaveButton;
                private JButton settingASHPrintButton;
                //Rema
                  private JLabel settingRELabel;
                  private JTextField settingRETextfield;
                        private JButton settingRESaveButton;
                        private JButton settingREPrintButton;
         public SettingsView() {
             frame = new JFrame("Settings");
                  frame.getContentPane().setLayout(new BorderLayout());
                  frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
                  frame.setSize(500, 120);
                  frame.setLocationRelativeTo(null);
                  frame.setVisible(true);

                   // Create UI elements/shahad
                  settingShLabel = new JLabel("shahad Setting :");
                  settingShTextfield = new JTextField();
                  settingShSaveButton = new JButton("Save shahad setting ");
                  settingShPrintButton = new JButton("Print to Console!");

                  // Create UI elements/Ashwaq
                  settingASHLabel = new JLabel("Ashwaq Setting :");
                  settingASHTextfield = new JTextField();
                  settingASHSaveButton = new JButton("Save Ashwaq setting ");
                  settingASHPrintButton = new JButton("Print to Console!");

                  // Create UI elements/Rema
                  settingRELabel = new JLabel("Rema Setting :");
                  settingRETextfield = new JTextField();
                  settingRESaveButton = new JButton("Save Rema setting ");
                  settingREPrintButton = new JButton("Print to Console!");

                   // Add UI element to frame
                  GroupLayout layout = new GroupLayout(frame.getContentPane());
                  layout.setAutoCreateGaps(true);
                  layout.setAutoCreateContainerGaps(true);
                  layout.setHorizontalGroup(layout.createSequentialGroup()
```

```
.addGroup(layout.createParallelGroup(GroupLayout.Alignment.LEADING).addComponent
(settingShLabel).addComponent(settingASHLabel).addComponent(settingRELabel))

.addGroup(layout.createParallelGroup(GroupLayout.Alignment.LEADING).addComponent
(settingShTextfield).addComponent(settingASHTextfield).addComponent(settingRETextfield))

.addGroup(layout.createParallelGroup(GroupLayout.Alignment.LEADING).addComponent
(settingShSaveButton).addComponent(settingASHSaveButton).addComponent(settingRESaveButton))

.addGroup(layout.createParallelGroup(GroupLayout.Alignment.LEADING).addComponent
(settingShPrintButton).addComponent(settingASHPrintButton).addComponent(settingREPrintButton)
));
                        layout.setVerticalGroup(layout.createSequentialGroup()

.addGroup(layout.createParallelGroup(GroupLayout.Alignment.BASELINE).addComponent
(settingShLabel).addComponent(settingShTextfield).addComponent(settingShSaveButton).
addComponent(settingShPrintButton))
.addGroup(layout.createParallelGroup(GroupLayout.Alignment.BASELINE).addComponent
(settingASHLabel).addComponent(settingASHTextfield).addComponent(settingASHSaveButton).
addComponent(settingASHPrintButton))
.addGroup(layout.createParallelGroup(GroupLayout.Alignment.BASELINE).addComponent
(settingRELabel).addComponent(settingRETextfield).addComponent(settingRESaveButton).
addComponent(settingREPrintButton)));
        frame.getContentPane().setLayout(layout);}
            public JFrame getFrame() {
             return frame;}
            public void setFrame(JFrame frame) {
             this.frame = frame;}

//shahad
            public JLabel getSettingShLabel() {
                        return settingShLabel;}
            public void setSettingShLabel(JLabel settingShLabel) {
                        this.settingShLabel = settingShLabel;}
             public JTextField getSettingShTextfield() {
                return settingShTextfield;}
            public void setSettingShTextfield(JTextField settingShTextfield) {
                this.settingShTextfield = settingShTextfield; }
    public JButton getSettingShSaveButton() {
                return settingShSaveButton;  }
            public void setSettingShSaveButton(JButton settingShSaveButton) {
                this.settingShSaveButton = settingShSaveButton; }
            public JButton getSettingShPrintButton() {
               return settingShPrintButton; }
            public void setShPrintButton(JButton print) {
                        this.settingShPrintButton = print; }
```

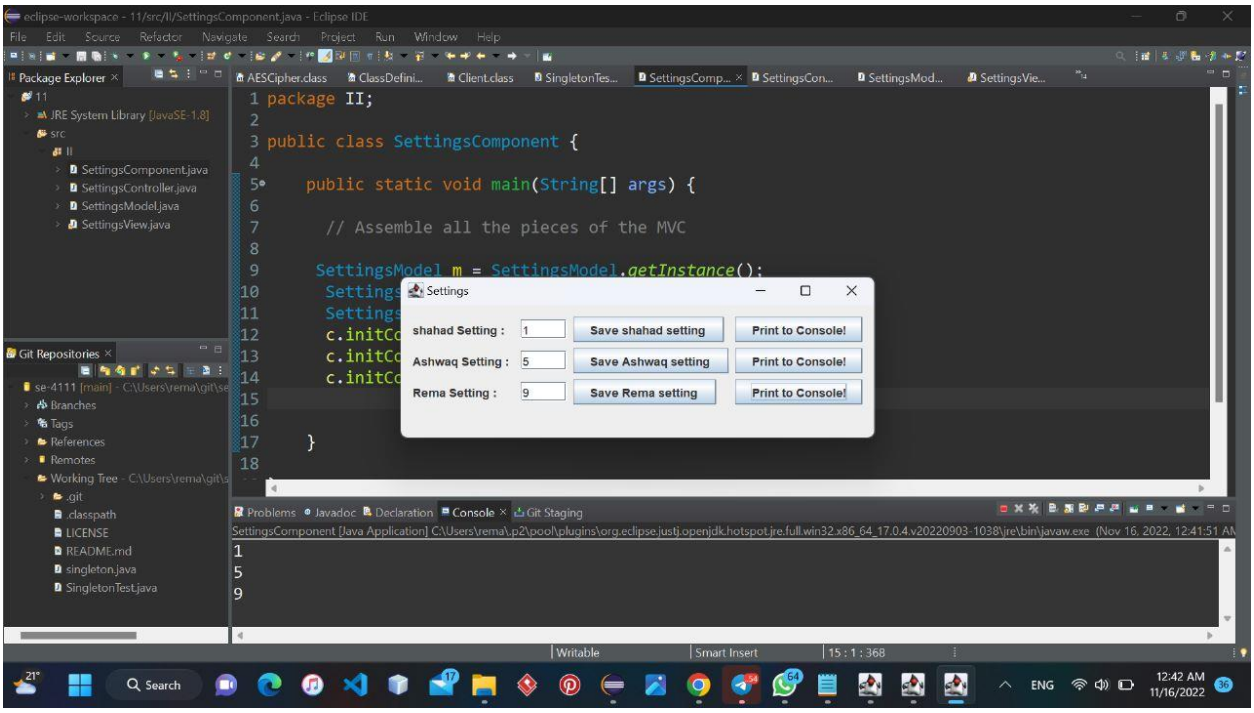
```
//Ashwaq
   public JLabel getSettingASHLabel() {
            return settingASHLabel;}
   public void setSettingASHLabel(JLabel settingASHLabel) {
            this.settingASHLabel = settingASHLabel;}
   public JTextField getSettingASHTextfield() {
     return settingASHTextfield; }
     public void setSettingASHTextfield(JTextField settingASHTextfield) {
         this.settingASHTextfield = settingASHTextfield;}
     public JButton getSettingASHSaveButton() {
         return settingASHSaveButton;  }
     public void setSettingASHSaveButton(JButton settingASHSaveButton) {
         this.settingASHSaveButton = settingASHSaveButton;}
     public JButton getSettingASHPrintButton() {
         return settingASHPrintButton;}
     public void setASHPrintButton(JButton print) {
                this.settingASHPrintButton = print; }

 //Rema
                    public JLabel getSettingRELabel() {
                              return settingShLabel; }
                   public void setSettingRELabel(JLabel settingRELabel) {
                              this.settingRELabel = settingRELabel;}
                    public JTextField getSettingRETextfield() {
                       return settingRETextfield; }
                    public void setSettingRETextfield(JTextField settingRETextfield) {
                       this.settingRETextfield = settingRETextfield; }
      public JButton getSettingRESaveButton() {
                       return settingRESaveButton; }

                    public void setSettingRESaveButton(JButton settingRESaveButton) {
                             this.settingRETextfield = settingRETextfield;}
                    public JButton getSettingREPrintButton() {
                      return settingREPrintButton; }
                    public void setREPrintButton(JButton print) {
                              this.settingREPrintButton = print; } }
```

3.3 User interface

-Before adding our settings:

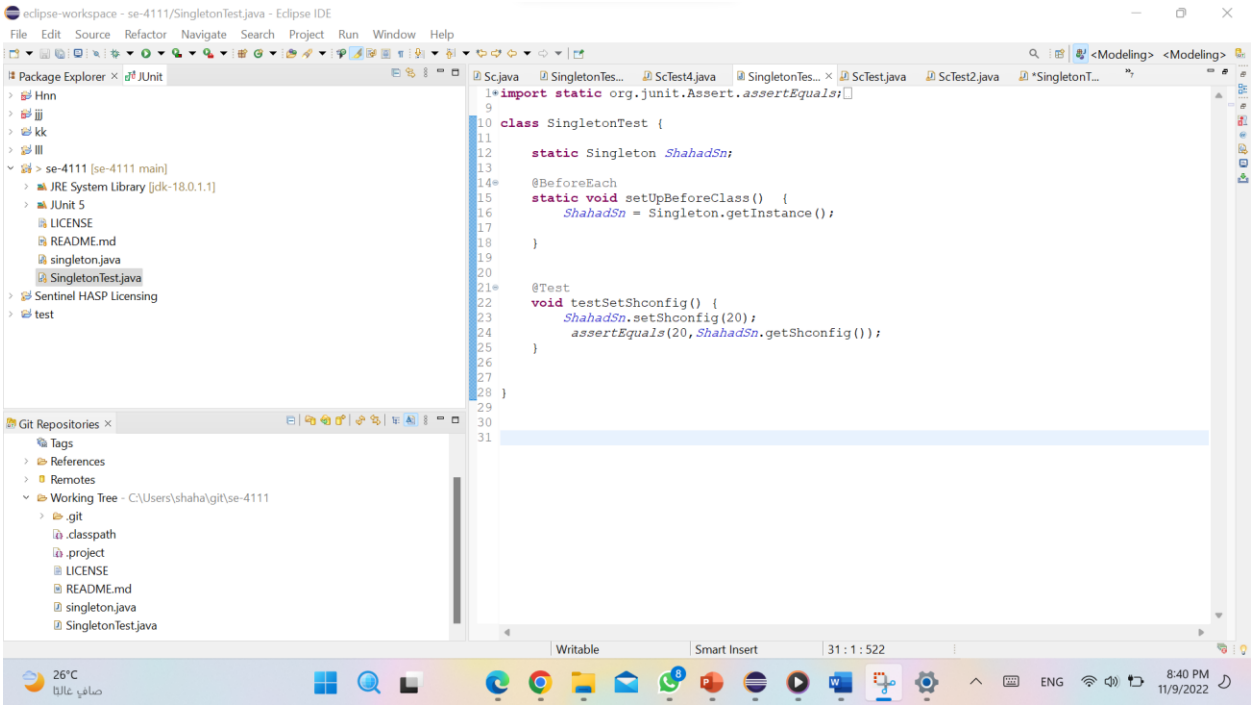


-After adding our settings:

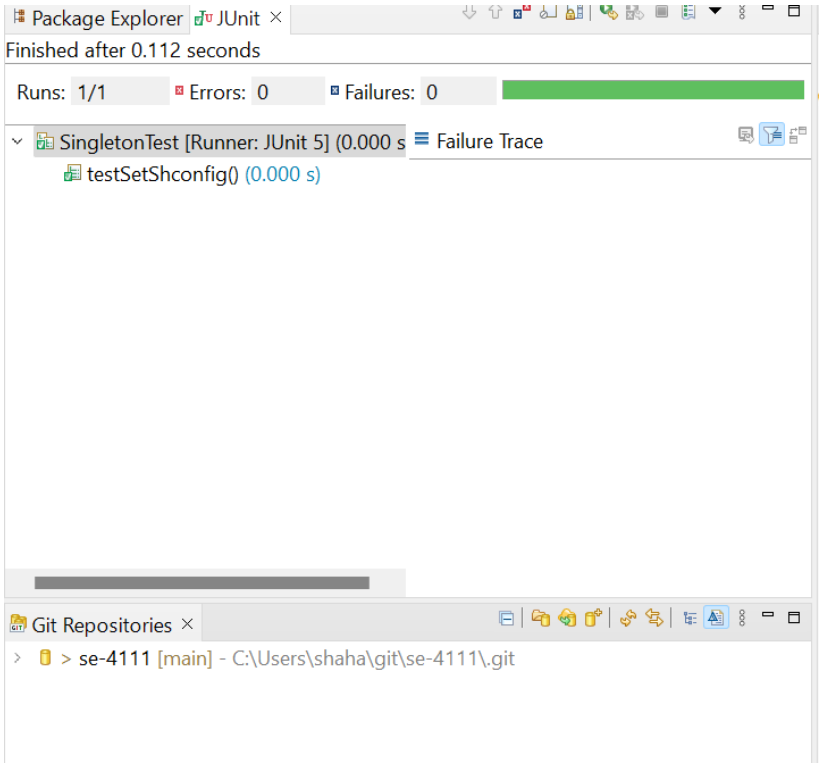


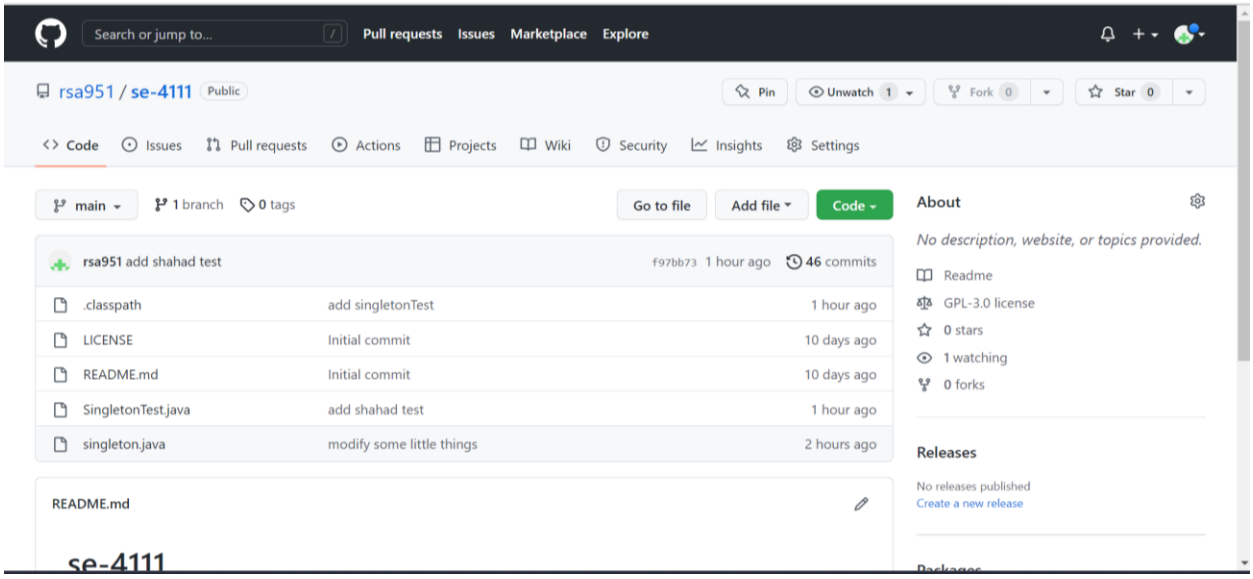
4. Testing

Shahad's test code:

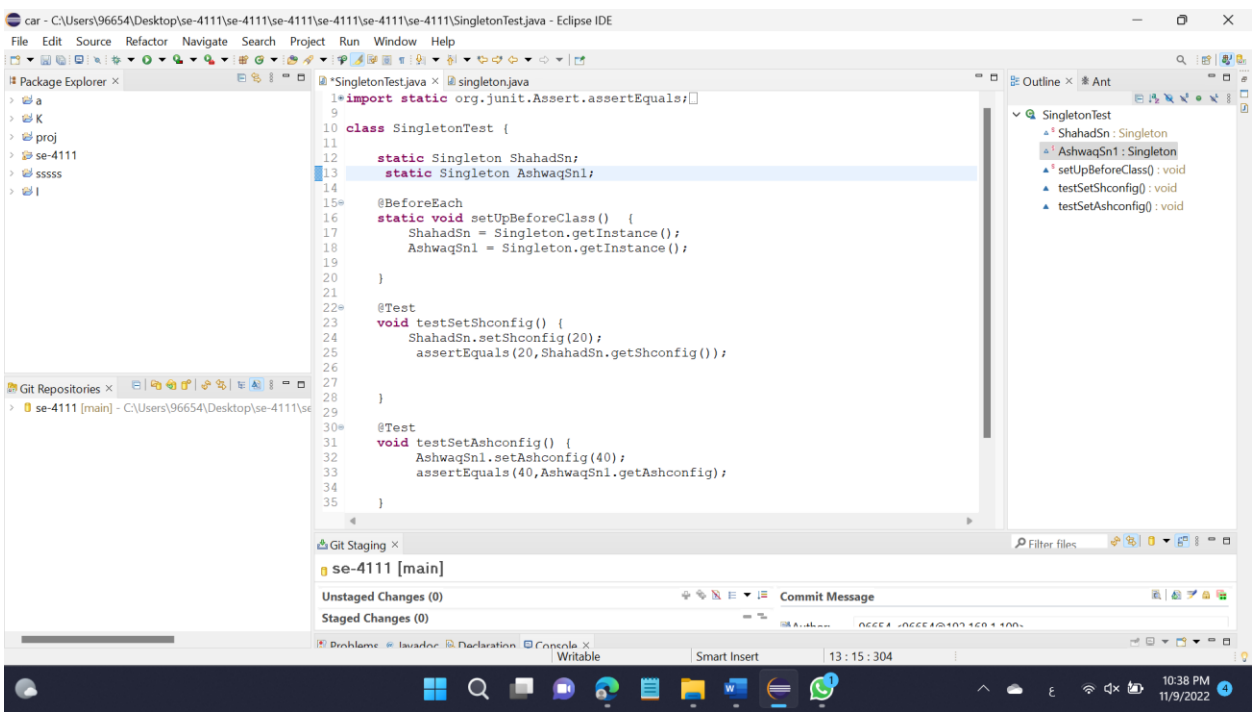


The result:

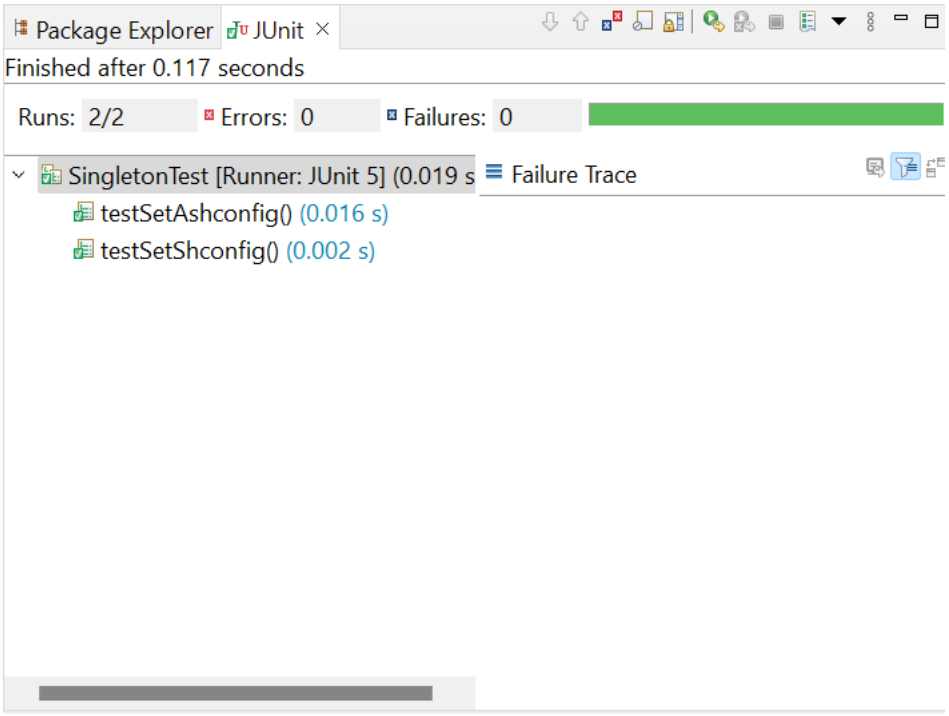


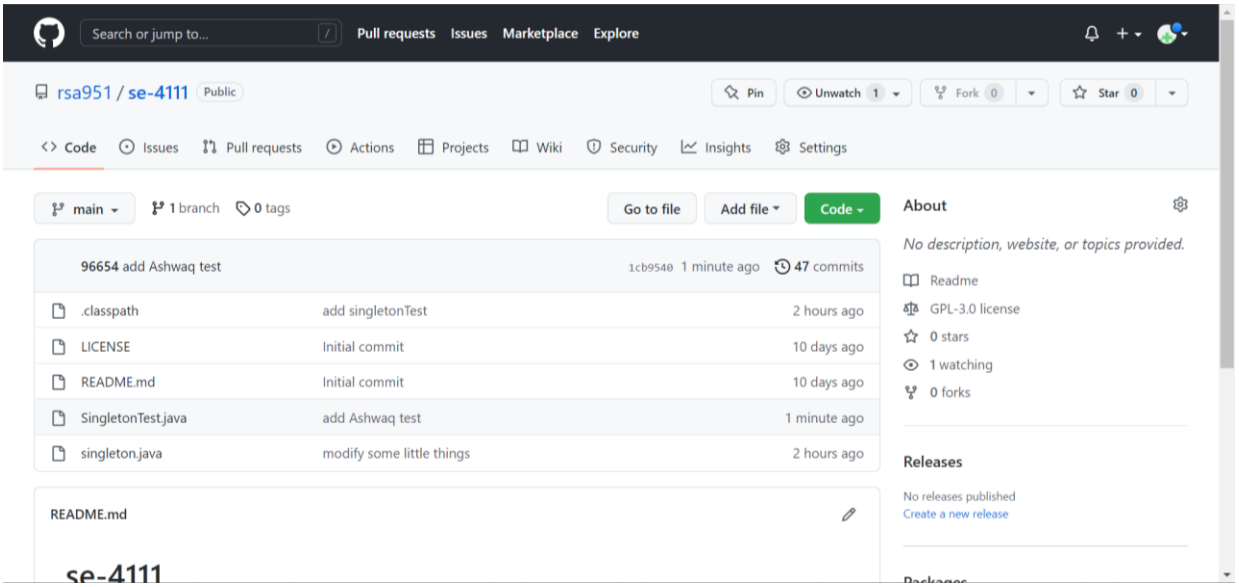


Ashwaq ’s test code:

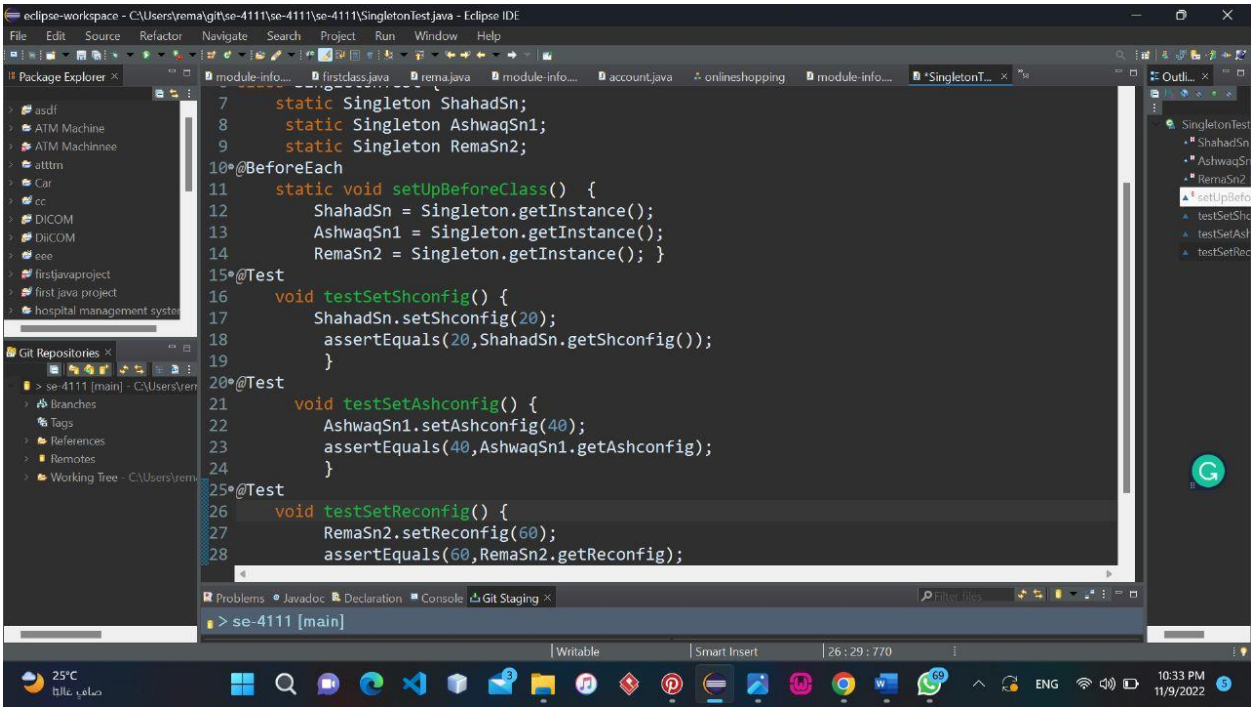


The result:

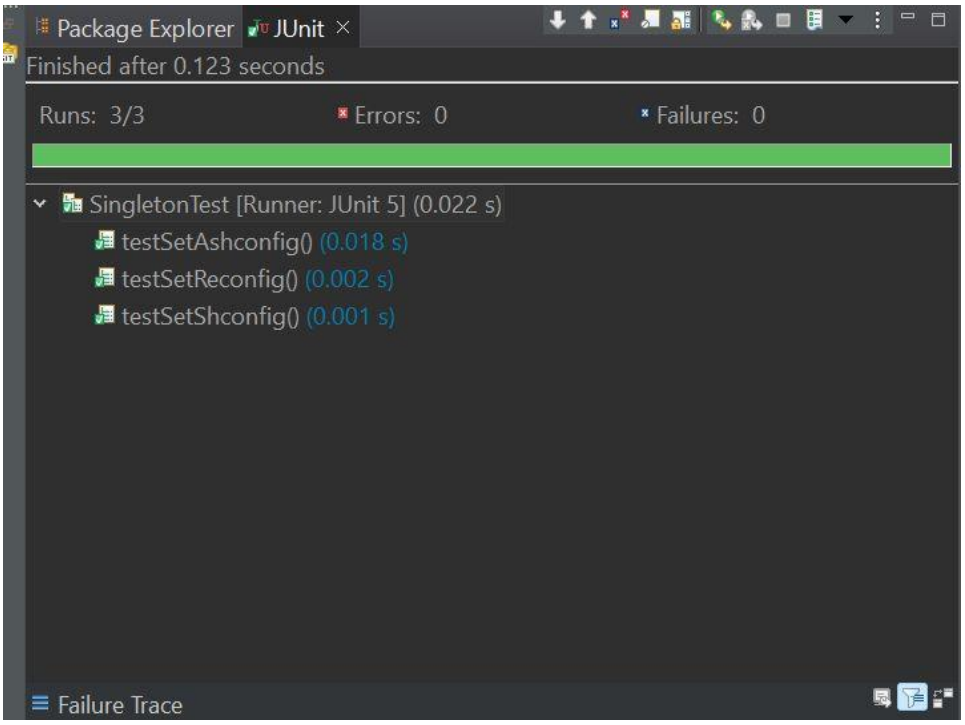

Package Explorer
JUnit
Finished after 0.117 seconds
Runs: 2/2
Errors: 0
Failures: 0
SingletonTest [Runner: JUnit 5] (0.019 s
Failure Trace
testSetAshconfig() (0.016 s)
testSetShconfig() (0.002 s)

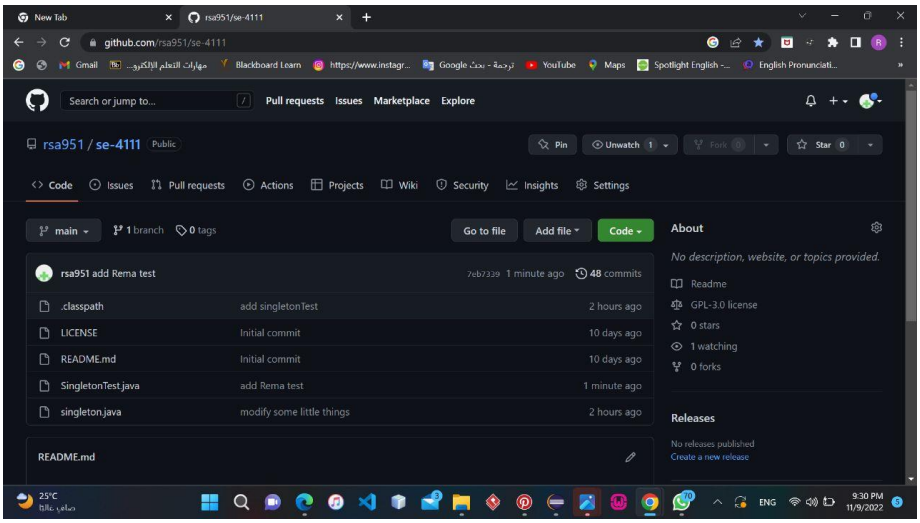

Search or jump to...
Pull requests
Issues
Marketplace
Explore
rsa951 / se-4111
Public
Pin
Unwatch 1
Fork 0
Star 0
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
main
1 branch
0 tags
Go to file
Add file
Code
96654 add Ashwaq test
1cb9540 1 minute ago
47 commits
.classpath
add singletonTest
2 hours ago
LICENSE
Initial commit
10 days ago
README.md
Initial commit
10 days ago
SingletonTest.java
add Ashwaq test
1 minute ago
singleton.java
modify some little things
2 hours ago
README.md
se-4111
About
No description, website, or topics provided.
Readme
GPL-3.0 license
0 stars
1 watching
0 forks
Releases
No releases published
Create a new release

Rema’s test code:



The result:





The code :

```
import static org.junit.Assert.assertEquals;
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.BeforeAll;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
class SingletonTest {
        static Singleton ShahadSn;
         static Singleton AshwaqSn1;
         static Singleton RemaSn2;
@BeforeEach
        static void setUpBeforeClass()  {
                ShahadSn = Singleton.getInstance();
                AshwaqSn1 = Singleton.getInstance();
                RemaSn2 = Singleton.getInstance(); }
@Test
        void testSetShconfig() {
                ShahadSn.setShconfig(20);
                 assertEquals(20,ShahadSn.getShconfig());}
@Test
  void testSetAshconfig() {
                 AshwaqSn1.setAshconfig(40);
                 assertEquals(40,AshwaqSn1.getAshconfig);}
@Test
void testSetReconfig() {
         RemaSn2.setReconfig(60);
                 assertEquals(60,RemaSn2.getReconfig);}}
```

5. Lessons learned

First, we learned how to take advantage of the GitHub , where we created a repository that can connect to Eclipse by means of push and pull operations, as it makes it easier for us to make modifications at any time and share it also it facilitates teamwork. And we learned how to write the code for a Singleton class with its setter and getter and how to write specification of functions and we learned how to run tests on methods and how to write test cases and finally we learned how to create an MCV model where we generated a settings.